DATA MINING COURSE

Topic 7 — Bank Management System

Exploratory Data Analysis Report

All 20 Questions — With Code, Results & Visualizations

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Introduction

This report documents the full Exploratory Data Analysis (EDA) and machine learning pipeline applied to a bank marketing campaign dataset. Each step of the workflow is presented as a question, explaining the requirements, actions taken, and key results observed.

The primary objective is to predict whether a client will subscribe to a term deposit based on their demographic and behavioural attributes.

1. Dataset Overview

Q1: What does the dataset look like, and does it contain missing or unknown values?

Requirements

Before analysis, we must understand the structure of the dataset: the number of entries, features, data types, and the presence of any missing or 'unknown' values that may affect modeling.

Actions Taken

Dataset loaded: The CSV file was read using pandas (separator = ';').

Shape & types: df.shape, df.info(), and df.describe() were used to inspect dimensions and data types.

Missing values: NaN values were checked in numerical columns; 'unknown' string values were counted in categorical columns.

Results

Metric	Value
Total Entries	41,188
Total Features	21 (including target)
Numerical Features	10
Categorical Features	11
Standard NaN Values	None found

'Unknown' values were found in several categorical columns:

Feature	'Unknown' Count	% of Total
poutcome	~36,959	89.66%
contact	~12,720	30.88%
education	~1,169	2.84%
job	~160	0.39%

These 'unknown' values are retained as valid categories during preprocessing rather than being imputed or removed.

2. Target Variable Distribution

Q2: Is the target variable balanced, and how does class imbalance affect modeling?

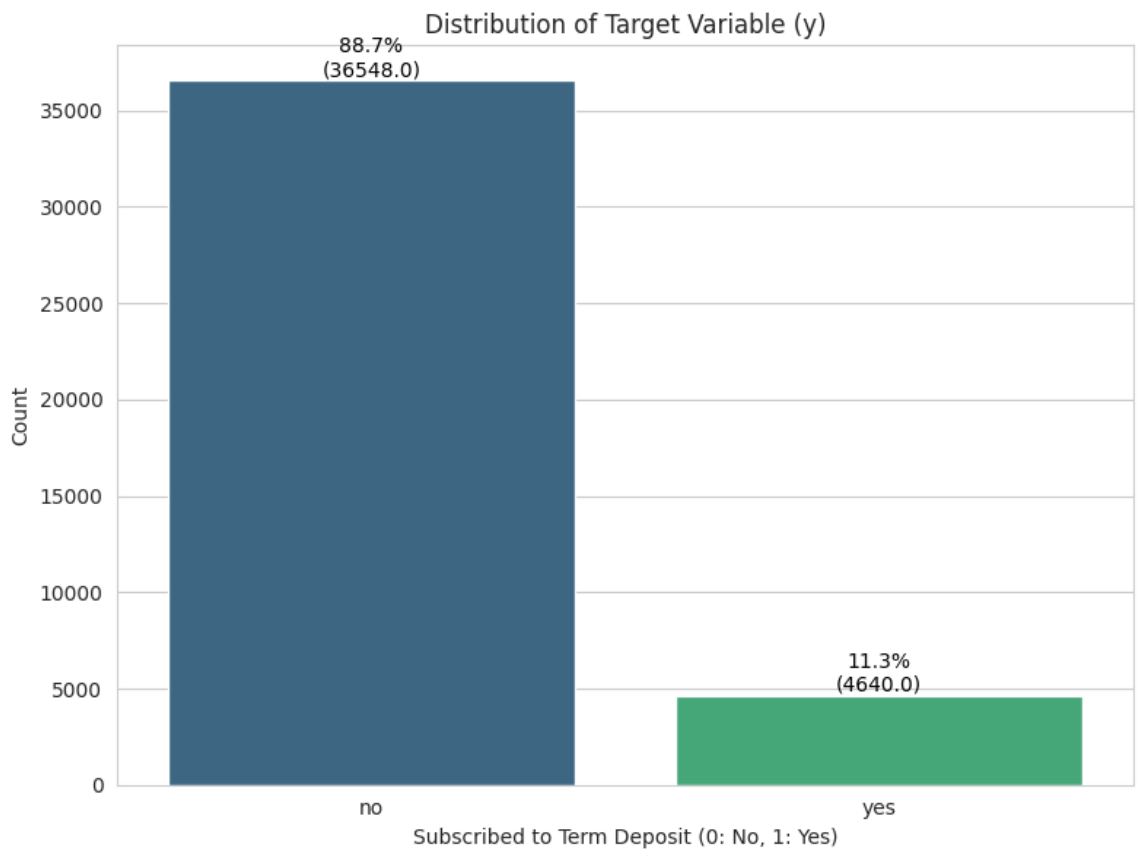
Requirements

Understanding the distribution of the target variable 'y' (subscription to term deposit: yes/no) is critical for choosing the right modeling strategy.

Actions Taken

A count plot was generated to visualize the frequency of each class, annotated with both the raw count and percentage.

Results



Class	Count	Percentage
No (0) – Did not subscribe	~36,548	~88.7%
Yes (1) – Subscribed	~4,640	~11.3%

The dataset is significantly imbalanced. This must be addressed during model training via techniques such as class weighting, oversampling (SMOTE), or threshold tuning to avoid a biased classifier.

3. Univariate Analysis

Q3: What are the individual distributions of numerical and categorical features?

Requirements

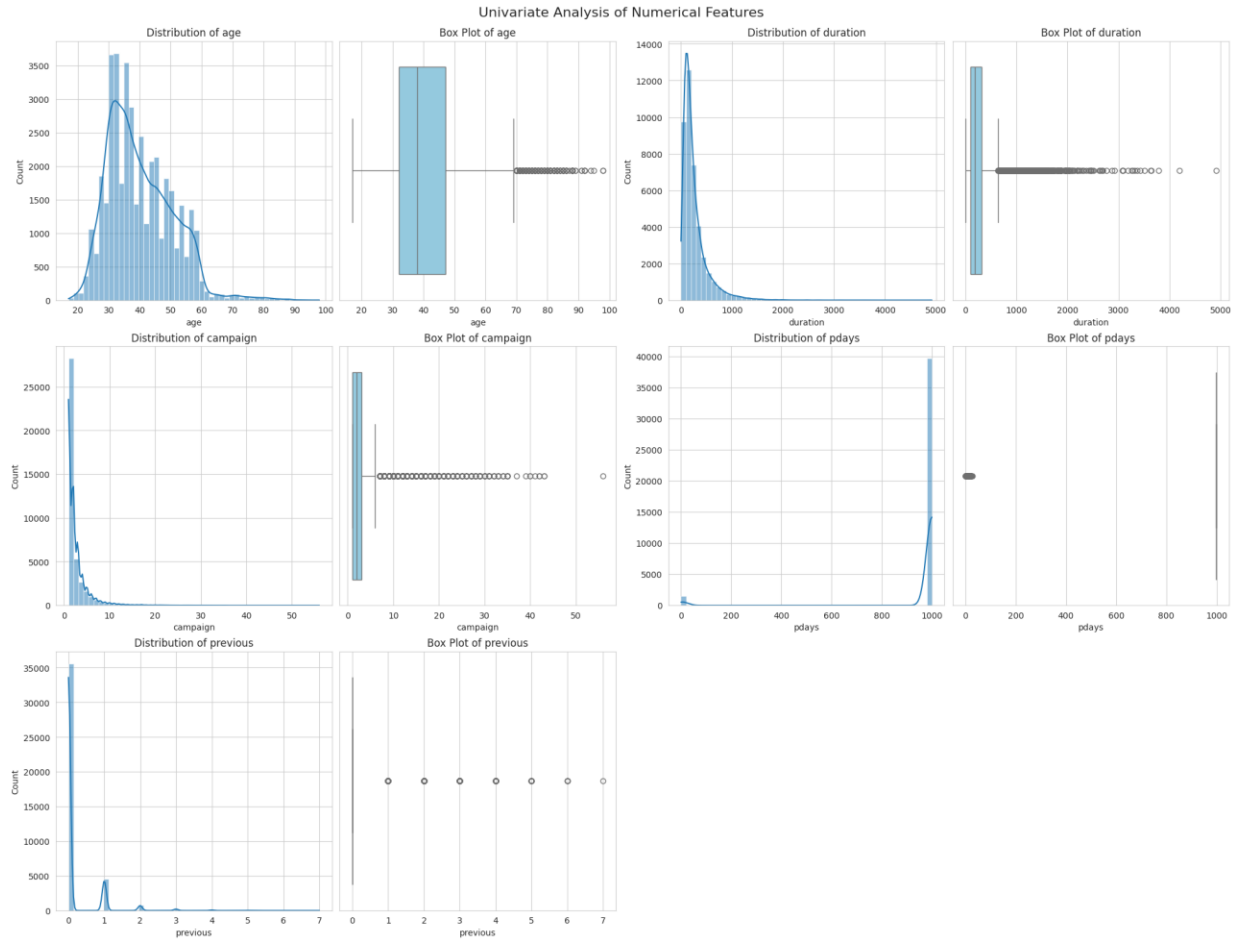
Univariate analysis examines each feature in isolation to detect skewness, outliers, and dominant categories — insights essential for preprocessing decisions.

3a. Numerical Features

Actions Taken

For each of the 5 core numerical features (age, duration, campaign, pdays, previous), histograms with KDE overlays and box plots were generated side-by-side.

Results



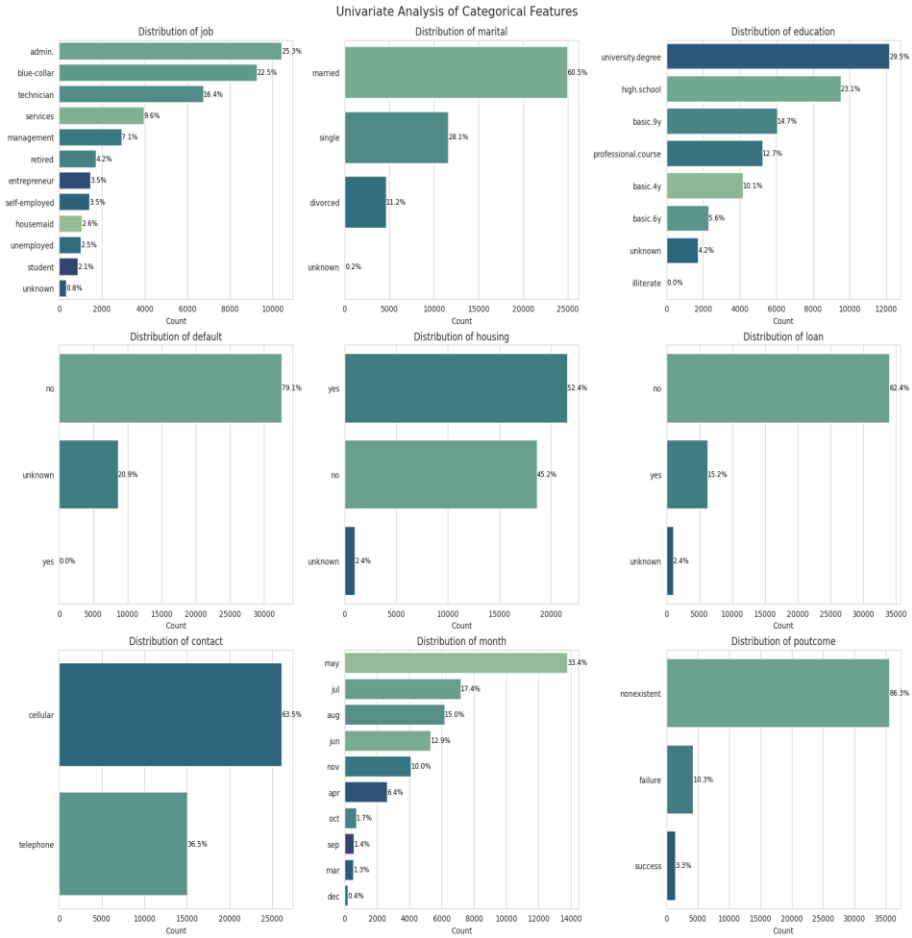
Feature	Distribution Type	Key Observation
age	Near-normal, slight right skew	Majority between 25–60 years
duration	Heavy right skew	Most calls short; some very long (outliers)
campaign	Strong right skew	Most clients contacted 1–3 times
pdays	Bimodal	Large spike at -1 (never contacted before)
previous	Extreme right skew	Most clients had 0 previous contacts

3b. Categorical Features

Actions Taken

Horizontal count bar plots were generated for 9 categorical features, with percentage annotations to highlight dominant categories.

Results



Feature	Dominant Category
job	admin., blue-collar, technician
marital	married (~60%)
education	university.degree, high.school
default	no (overwhelming majority)
housing	yes (~52%)
loan	no (~82%)
contact	cellular (~63%)
month	may (highest campaign activity)
poutcome	nonexistent (~89%)

4. Bivariate Analysis

Q4: How do individual features relate to the target variable (subscription outcome)?

Requirements

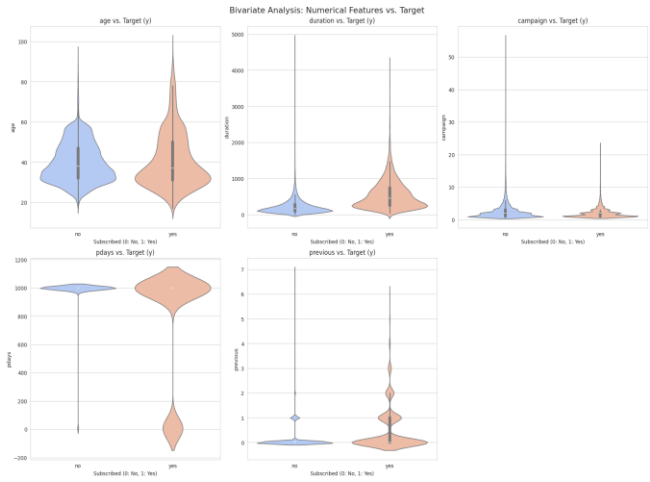
Bivariate analysis reveals which features are most predictive of term deposit subscription, guiding feature selection for the model.

4a. Numerical Features vs. Target

Actions Taken

Violin plots were used to compare the distribution of each numerical feature across the two target classes (subscribed vs. not subscribed).

Results



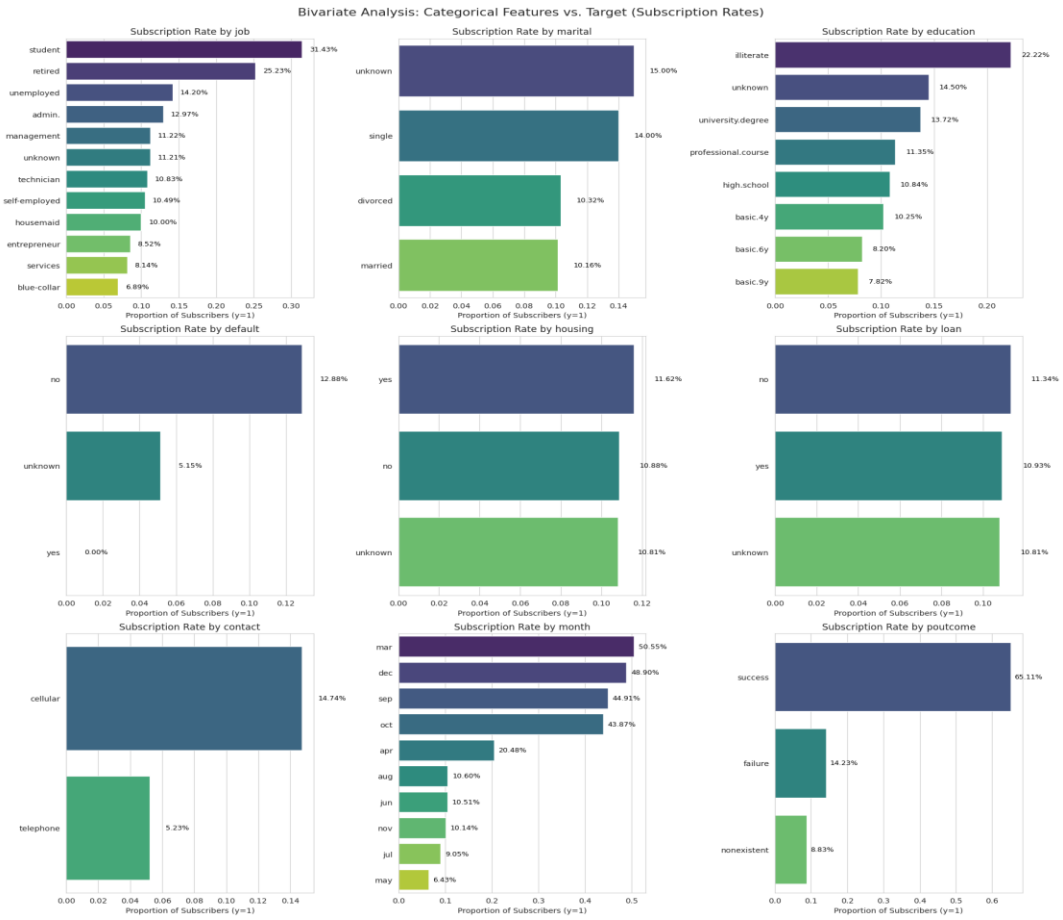
Feature	Finding
duration	Strongly separates classes — longer calls → higher subscription rate
previous	Subscribers had slightly more prior contacts
pdays	Subscribers more likely to have been contacted before (pdays ≠ -1)
campaign	No strong separation; subscribers contacted slightly fewer times
age	Slight tendency for older clients to subscribe

4b. Categorical Features vs. Target

Actions Taken

Grouped bar plots showed the proportion of 'yes' (subscription) within each category, sorted by subscription rate.

Results



Feature	Insight
poutcome	Previous success → very high subscription rate (~65%)
contact	Cellular contact → higher subscription rate than telephone
month	March, December, September show higher subscription rates
education	Illiterate and university-degree clients show higher rates
job	Retired and student clients subscribe at higher rates
marital	Single clients subscribe at slightly higher rates

5. Correlation Analysis

Q5: Are there significant correlations between numerical features, and do any features correlate with the target?

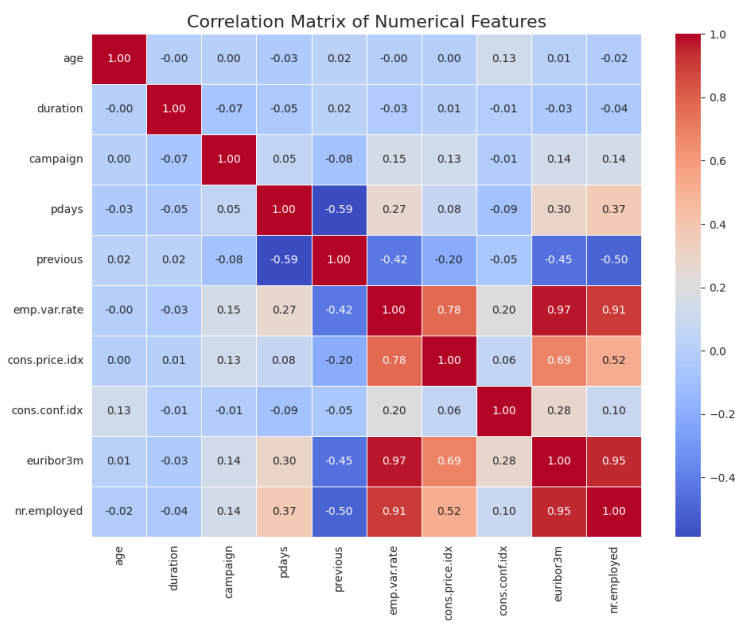
Requirements

Correlation analysis helps identify multicollinearity between features (which can harm model interpretability) and shows which numerical variables are most associated with the target.

Actions Taken

A Pearson correlation heatmap was generated for 10 numerical features plus the binary target 'y'.

Results



Feature Pair / Feature vs Target	Correlation	Interpretation
emp.var.rate ↔ euribor3m	~0.97	Very high multicollinearity
emp.var.rate ↔ nr.employed	~0.91	High multicollinearity
euribor3m ↔ nr.employed	~0.95	High multicollinearity
duration → y	~0.40	Strongest positive predictor
euribor3m → y	~-0.31	Negative correlation with subscription
emp.var.rate → y	~-0.30	Negative correlation
pdays → y	~0.10	Weak positive association

The high multicollinearity among macroeconomic indicators (emp.var.rate, euribor3m, nr.employed) may necessitate dimensionality reduction or careful feature selection.

6. Data Preprocessing & Feature Engineering

Q6: How was the raw data cleaned, transformed, and enriched to prepare it for modeling?

Requirements

Raw categorical data must be encoded numerically, skewed distributions should be normalized, and new features can be engineered to capture domain knowledge — all before feeding data to a machine learning model.

Actions Taken

Step	Action	Detail
1	Drop 'id' column	Removed non-informative identifier (if present)
2	One-Hot Encoding	All object-type columns encoded via pd.get_dummies()
3	Log1p transformation	Applied to duration, campaign, previous (right-skewed)
4	pdays special handling	log1p(pdays + 2) to handle -1 values before log
5	Feature engineering	Created binary was_contacted_before (pdays ≠ -1)
6	Target encoding	y mapped from {'no':0, 'yes':1}

Results

```
Shape before preprocessing : (41188, 21)
Shape after preprocessing  : (41188, 65)
New engineered feature     : was_contacted_before
Target variable            : y → {0: no, 1: yes}
```

The dataset expanded from 21 to 65 columns after one-hot encoding of 9 categorical features. The was_contacted_before feature explicitly captures whether the client had any prior campaign contact.

7. Feature Scaling

Q7: Why and how were numerical features standardized?

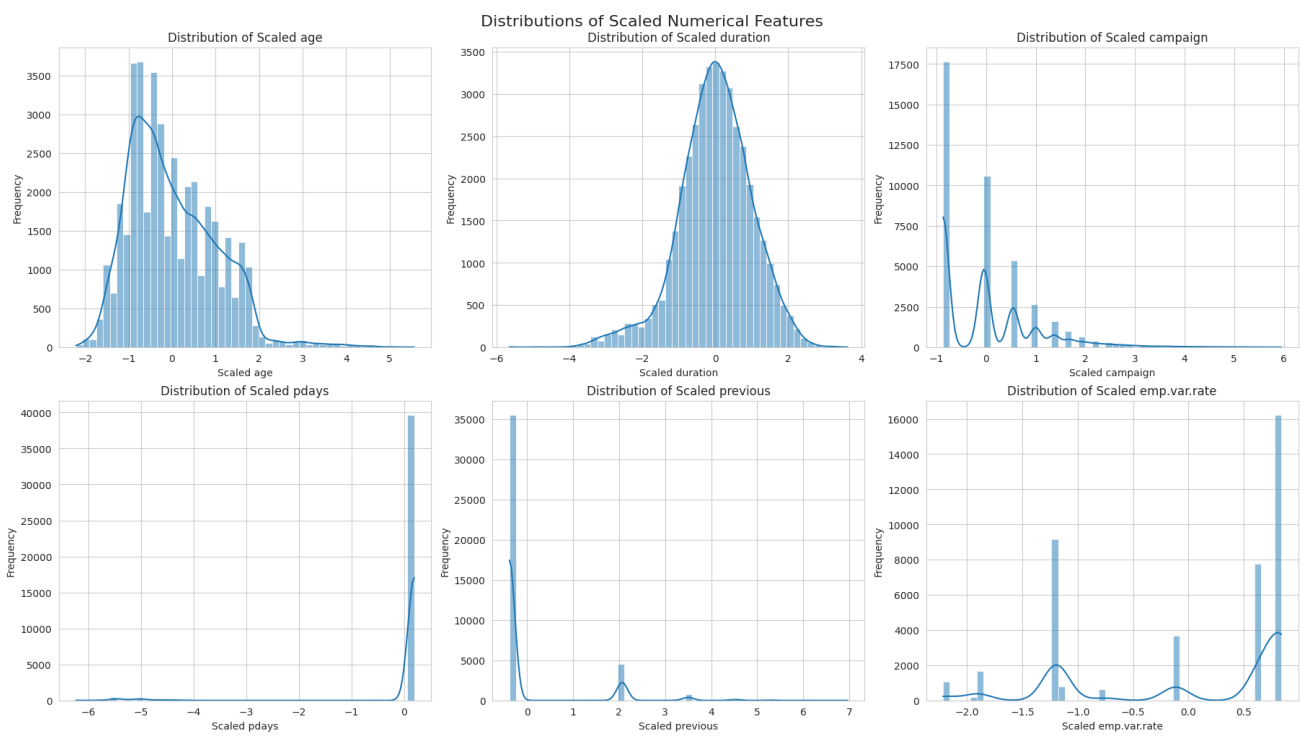
Requirements

Many machine learning algorithms (e.g., logistic regression, SVMs, neural networks) are sensitive to the scale of input features. Standardization ensures all features contribute equally.

Actions Taken

StandardScaler (zero mean, unit variance) was applied to 10 numerical features: age, duration, campaign, pdays, previous, emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m, nr.employed.

Results




```
Shape after scaling: (41188, 65)  [unchanged]
All scaled features: mean ≈ 0, std ≈ 1
```

Post-scaling distributions are centred around zero with unit variance. Features such as duration and campaign that were previously highly right-skewed are now more compact and comparable in scale.

Summary

The following table provides a concise overview of the entire analysis pipeline:

#	Process	Key Result
1	Dataset Overview	41,188 rows × 21 features; 'unknown' values handled as valid categories
2	Target Distribution	88.7% No / 11.3% Yes — significant class imbalance
3	Univariate Analysis	Numerical features right-skewed; categorical features show dominant classes
4	Bivariate Analysis	duration, poutcome, and contact type are strongest predictors
5	Correlation Analysis	High multicollinearity among macroeconomic features; duration best predictor
6	Preprocessing	One-hot encoding + log1p transforms; dataset expanded to 65 features